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**AI-Powered Real-Time Speech-to-Speech Video Translation System Using
Automatic Speech Recognition and Natural Language Processing**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of

DSA0308-Natural Language Processing

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BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE ENGINEERING WITH ARTIFICIAL
INTELLIGENCE**

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**AI-Powered Real-Time Speech-to-Speech Video Translation System Using Automatic Speech Recognition and Natural Language Processing**" has been carried out by **AB Dhanyashree (192472002)** under the supervision of **Dr K Kumaragurubaran** and is submitted in partial fulfilment of the requirements for the current semester of the B.E CSE-AI program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **AB Dhanyashree (192472002)** of the Computer Science Engineering Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**AI-Powered Real-Time Speech-to-Speech Video Translation System Using Automatic Speech Recognition and Natural Language Processing**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 08-09-2026

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

The contribution of each team member towards the design, development, and documentation of this Capstone Project is summarised below.

Student Name (Reg. No.)	Primary Area of Contribution	Approx. Share (%)
Keerthana Sai K (192425143)	Automatic Speech Recognition module — audio capture, pre-processing, and integration of the speech-to-text engine; testing and documentation of Chapters 1 and 4.	34%
Sunayana P (192424303)	Natural Language Processing / Machine Translation module — text segmentation, translation pipeline, and evaluation; system architecture and documentation of Chapters 2 and 3.	33%
A B Dhanya Shree	Text-to-Speech synthesis, video/audio synchronisation, and user-interface integration; testing, results compilation, and documentation of Chapter 5 and Appendices.	33%

ABSTRACT

The AI-Powered Real-Time Speech-to-Speech Video Translation System is an intelligent multimedia application developed to break down language barriers in video communication by automatically converting spoken content in a source language into natural-sounding speech in a target language, while keeping the translated audio synchronised with the original video. Manually subtitling, dubbing, or interpreting video content is slow, expensive.

The proposed system combines Automatic Speech Recognition (ASR), Natural Language Processing (NLP)-based Neural Machine Translation (NMT), and Text-to-Speech (TTS) synthesis into a single real-time pipeline. Audio extracted from an incoming video stream is segmented using voice-activity detection and transcribed using a deep-learning ASR engine; the resulting text is cleaned, segmented into meaningful chunks, and translated into the target language using a transformer-based NMT model; the translated text is then synthesised into speech using a neural TTS engine and re-mixed with the original video, with timing adjustments applied to keep the dubbed audio aligned to the speaker's utterances.

The methodology followed a systematic pipeline of requirement analysis, data collection, model selection, module integration, latency optimisation, and evaluation, implemented primarily in Python using open-source libraries such as Whisper/SpeechRecognition for ASR, Hugging Face Transformers (MarianMT/NLLB) for translation, and neural TTS toolkits (e.g., Coqui TTS) for speech synthesis, together with FFmpeg for audio/video processing. The system was evaluated on transcription accuracy (Word Error Rate), translation quality (BLEU score), speech naturalness (Mean Opinion Score), and end-to-end latency, demonstrating that an integrated ASR–NLP–TTS pipeline can deliver understandable, near-real-time translated speech for video content while operating within practical latency and resource constraints.

The platform is intended as a communication-assistance and accessibility tool rather than a certified professional interpretation service, and careful handling of audio/video data, user privacy, and consent remains an essential part of responsible deployment. The project concludes that combining speech recognition, language translation, and speech synthesis in a single pipeline can meaningfully reduce the language barrier in video-based communication .

Keywords: Automatic Speech Recognition, Natural Language Processing, Neural Machine Translation, Text-to-Speech Synthesis, Real-Time Video Translation, Speech-to-Speech Translation, Deep Learning

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
NLP	Natural Language Processing	-
ML	Machine Learning	-
AI	Artificial Intelligence	-
F1	F1-score (harmonic mean of precision and recall)	-

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Video has become the dominant medium for education, business communication, entertainment, and social interaction, and an increasing share of this content is produced and consumed across language boundaries — through video calls, recorded lectures, webinars, and streamed broadcasts. Understanding spoken content in an unfamiliar language traditionally relies on human interpreters, professional dubbing studios, or subtitle translators, all of which are time-consuming, costly, and difficult to apply to live or near-live video.

The AI-Powered Real-Time Speech-to-Speech Video Translation System is proposed as an intelligent solution to this problem. It combines Automatic Speech Recognition (ASR), Natural Language Processing (NLP)-based machine translation, and Text-to-Speech (TTS) synthesis to automatically convert the spoken audio of a video from a source language into natural-sounding speech in a target language, while keeping the generated audio aligned with the original video timeline.

1.2 Need for the Project

With the growth of global video communication — cross-border meetings, international online courses, multilingual social content, and live streaming — the inability to understand spoken content in another language remains a persistent barrier. Users are affected in different ways: professionals need real-time understanding of multilingual meetings without waiting for a human interpreter; content creators and educators need an affordable way to localise video content into multiple languages; and viewers of live broadcasts need near-instant translated audio rather than static subtitles that can be hard to read on small screens or while multitasking. Existing subtitle-only translation tools are limited because they require the viewer to read rather than listen, and they do not reproduce natural speech, tone, or timing. This represents a genuine engineering gap between the growing volume of cross-lingual video content and the tools available to make it accessible in real time.

1.3 Problem Statement

Video communication increasingly crosses language boundaries, yet existing solutions — human interpretation, studio dubbing, and static subtitles — cannot scale to the volume, cost, and latency requirements of everyday and live video use. This combination of noisy,

continuous audio streams, language-pair variability, the need for natural and intelligible synthesised speech, and strict latency constraints for near-real-time use creates a multi-factor engineering problem: an effective solution must simultaneously recognise speech accurately under real-world acoustic conditions, translate the recognised text with adequate fluency and meaning preservation, synthesise natural target-language speech, and keep the translated audio synchronised with the source video — all under the constraint that the pipeline must run within practical latency and computational limits, and must respect user privacy and consent when processing voice and video data.

1.4 Project Aim

To design and develop an intelligent, modular pipeline that performs real-time speech-to-speech translation of video content by integrating Automatic Speech Recognition, Natural Language Processing-based machine translation, and Text-to-Speech synthesis, producing intelligible, time-aligned translated audio for the source video.

1.5 Project Objectives

1. To design an Automatic Speech Recognition module that accurately transcribes spoken audio extracted from video into source-language text under real-world acoustic conditions.
2. To develop a Natural Language Processing-based translation module that converts the transcribed text into fluent, meaning-preserving text in the target language.
3. To model a Text-to-Speech synthesis module that converts translated text into natural-sounding target-language speech.
4. To implement a synchronisation mechanism that aligns the synthesised speech with the timing of the original video and re-muxes the translated audio with the video stream.
5. To evaluate the system using standard metrics — Word Error Rate (WER) for recognition, BLEU score for translation quality, Mean Opinion Score (MOS) for speech naturalness, and end-to-end latency.
6. To assess the overall system's usability, reliability, and effectiveness as a real-time video-translation and accessibility tool.

1.6 Scope

Included: extraction and segmentation of speech audio from video, automatic speech recognition, text pre-processing and neural machine translation, text-to-speech synthesis, audio-video synchronisation and re-muxing, and evaluation through accuracy, fluency, naturalness, and latency metrics.

Excluded: the platform does not provide certified/legal interpretation services, does not perform visual lip-synchronisation of the speaker's mouth movements, and does not replace human interpreters in high-stakes settings (e.g., legal or medical proceedings) where certified accuracy is mandated.

Assumptions: input videos contain reasonably clear, single-speaker-dominant speech; users have basic digital literacy to operate the application and select source/target languages.

Boundary conditions: the system operates on recorded video files and live audio/video streams with standard audio sampling rates; it does not process sign-language video or silent video without an audio track.

1.7 Expected Engineering Outcome

The expected outcome is a working software prototype — a Python-based pipeline integrating an ASR module, an NLP-driven neural translation module, and a TTS synthesis module, connected through an audio/video processing layer, and presented through a simple user interface that allows a user to upload or stream a video, select source and target languages, and receive a translated video/audio output.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, standards, and existing technologies were identified through review of official documentation of the core libraries and frameworks used (Python, PyTorch, Hugging Face Transformers, OpenAI Whisper, Coqui TTS, FFmpeg), together with foundational texts and research papers in automatic speech recognition, neural machine translation, and speech synthesis, as well as guidance published by the World Health Organization (WHO)-style accessibility frameworks and the NIST AI Risk Management Framework for the responsible deployment of AI systems that process voice data.

2.2 Review of Existing Work

Existing research in speech and language processing shows that deep-learning-based ASR systems (e.g., encoder-decoder and transformer architectures such as Whisper) have substantially reduced Word Error Rates compared with earlier statistical models, particularly under noisy or accented speech conditions. Standard references in Natural Language Processing describe tokenisation, sequence-to-sequence modelling, and attention mechanisms that underpin modern Neural Machine Translation systems, while research on encoder-based translation models (e.g., MarianMT, NLLB) documents their ability to translate across a very large number of language pairs with competitive BLEU scores. Literature on neural Text-to-Speech synthesis (e.g., Tacotron-style and vocoder-based architectures) describes how mel-spectrogram prediction followed by neural vocoding produces markedly more natural speech than earlier concatenative or parametric TTS. Existing commercial video-translation and captioning tools (subtitle generators, dubbing services, live-caption features in conferencing software) were also reviewed; most either provide only text subtitles or require significant post-production time for full dubbing, rather than an integrated, real-time speech-to-speech pipeline.

2.3 Comparative Analysis

Table 2.1 Advantages and Limitations of Existing Approaches

Approach	Advantages	Limitations
Human interpretation / studio dubbing	High accuracy and naturalness; handles nuance, tone, and context well	Expensive, slow, and impossible to scale to large volumes of everyday or live video

Approach	Advantages	Limitations
Static subtitle translation	Low computational cost; widely supported by video platforms	Requires reading rather than listening; not usable in audio-only or hands-free scenarios
Live captioning (ASR + text only)	Reasonably fast; useful for accessibility	No translation to another spoken language; still text-only output
Standalone machine translation apps (text-only)	Fast, widely available for many language pairs	Not integrated with audio/video; requires the user to manually transcribe and re-enter text
Proposed ASR + NLP + TTS speech-to-speech pipeline	Produces natural translated speech synchronised with the source video; usable for live and recorded content	Cumulative errors from ASR, translation, and TTS stages; requires careful latency and resource management

2.4 Research / Engineering Gap

Existing tools tend to address only one stage of the pipeline — recognition, translation, or synthesis — in isolation, or they trade real-time usability for post-production quality (as in professional dubbing). There remains a gap for an integrated, end-to-end pipeline that combines automatic speech recognition, NLP-based machine translation, and speech synthesis within a single accessible system capable of operating close to real time on recorded or streamed video, while remaining transparent about its accuracy limitations.

2.5 Proposed Contribution

This project contributes an integrated pipeline that combines ASR-based transcription, NLP-driven neural machine translation, and neural TTS synthesis in one system, with audio-video synchronisation and multi-metric evaluation (WER, BLEU, MOS, latency). Unlike isolated captioning or translation tools, the proposed system links recognition directly to translation and then to natural speech synthesis, reducing the manual effort otherwise required to transcribe, translate, and re-voice video content separately.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements — Stakeholder / User Requirements

Table 3.1 Stakeholder / User Requirements

Stakeholder	Requirement
Meeting participants / professionals	Near-real-time translated audio during video calls with minimal delay
Content creators / educators	Ability to translate recorded lecture or content videos into multiple target languages
General viewers	Intelligible, natural-sounding translated speech rather than text-only subtitles
System administrators	Clear language-pair coverage, error visibility, and resource/latency monitoring

Table 3.2 Functional & Non-Functional Requirements

Type	Requirement
Functional	Extract and segment speech audio from an uploaded or streamed video
Functional	Transcribe source-language speech to text using an ASR engine
Functional	Translate transcribed text into a user-selected target language
Functional	Synthesise translated text into natural target-language speech
Functional	Re-mux synthesised audio with the original video, preserving approximate timing
Non-Functional	End-to-end latency should remain within a few seconds per utterance for near-real-time use
Non-Functional	The system should support multiple source and target languages
Non-Functional	The system should degrade gracefully (e.g., fall back to captions) if synthesis confidence is low

3.2 Constraints & Applicable Standards

The system is constrained by available computational resources (CPU/GPU), the accuracy limitations of open-source ASR, NMT, and TTS models for certain language pairs, and the need to process audio in short, low-latency segments rather than waiting for an entire video. Relevant standards and guidance considered include general software-engineering testing practice, accessibility guidance for captioned/translated media, and the NIST AI Risk Management Framework for responsible AI deployment.

Table 3.3 Applications of the Project

Application Area	Description
Video conferencing	Real-time translated audio for cross-lingual meetings
E-learning	Automatic translation of recorded lectures for international students
Media localisation	Faster, lower-cost first-pass dubbing for streaming content
Accessibility	Spoken translation for viewers who cannot easily read subtitles

3.3 Design Specifications

Table 3.4 Project Target and Requirements

Metric	Target
Word Error Rate (ASR)	As low as practical for clear speech; benchmarked per language
Translation Quality (BLEU)	Competitive with open-source NMT baselines for supported language pairs
Speech Naturalness (MOS)	Rated on a standard 1–5 opinion scale by test listeners
End-to-end Latency	Minimised per utterance segment for near-real-time responsiveness

3.4 Alternative Solutions & Evaluation

Alternative 1 (Subtitle-only translation): Extract speech, transcribe, and translate to display text captions only. Simple and low-latency, but does not produce spoken output and requires the viewer to read.

Alternative 2 (Cascaded ASR + NLP + TTS pipeline — selected): Chains a speech recognition model, a neural machine translation model, and a neural text-to-speech model, with a synchronisation layer. More complex to build and tune, but delivers natural translated speech rather than text alone, and is substantially more usable for live or hands-free scenarios.

Table 3.5 Project Criteria and Weightage

Criterion	Weightage
Translation/transcription accuracy	35%
Speech naturalness of output	25%
End-to-end latency	25%
Ease of use / usability	15%

3.5 Selected Design & Detailed Design

The cascaded ASR + NLP + TTS pipeline (Alternative 2) was selected because it directly satisfies the functional requirements for producing spoken translated output, which subtitle-only translation cannot provide, and because it aligns with the engineering gap identified in Section 2.4.

Fig 3.5.1 Architecture Design

The selected design consists of six interfacing subsystems: (1) a User Interface for uploading/streaming video and selecting source/target languages; (2) an Audio Extraction & Segmentation module that separates the audio track and applies voice-activity detection to form utterance-level chunks; (3) a Speech Recognition module that transcribes each chunk into source-language text using a deep-learning ASR engine; (4) a Translation module that applies NLP-based Neural Machine Translation to convert the text into the target language; (5) a Speech Synthesis module that converts translated text into target-language audio using a neural TTS engine; and (6) a Synchronisation & Re-muxing module that time-aligns the synthesised audio with the original video and produces the final translated output. The Translation module depends on the output of the Speech Recognition module, and the Speech Synthesis module depends on the output of the Translation module, forming a sequential real-time pipeline.

3.6 Trade-off Analysis

Table 3.6 Designing and Planning

Trade-off	Decision
Accuracy vs. Latency	Use compact/optimised ASR and TTS model variants for near-real-time performance, with an option to switch to larger models for offline/high-accuracy processing of recorded video
Coverage vs. Quality	Prioritise a curated set of high-resource language pairs with strong NMT/TTS support over broad but lower-quality coverage
Segment length vs. Context	Use short, VAD-based utterance segments to reduce latency, accepting some loss of longer-range translation context

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.7 Domain and Evidence Evaluated

Domain	Consideration
Sustainability	Favour efficient, open-source models over large proprietary ones to reduce computational and energy footprint
Ethics	Obtain user consent before processing voice/video; avoid misrepresenting synthesised speech as the original speaker's authentic voice without disclosure
Safety	Clearly label translated/synthesised audio as AI-generated to prevent misuse (e.g., impersonation, misinformation)
Security	Process audio/video locally or over secure channels; avoid persistent storage of sensitive recordings beyond what is necessary

Risk Register

Risk	Likelihood	Impact	Mitigation
Mistranslation of critical content	Medium	High	Display disclaimers; allow user review/correction; avoid use in high-stakes settings
High latency under load	Medium	Medium	Use lightweight models, batching, and hardware acceleration where available
Voice-cloning / impersonation misuse	Low	High	Use generic/neutral synthetic voices by default; label output as AI-generated
Data privacy exposure	Low	High	Minimise retention of raw audio/video; encrypt data in transit and at rest

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The platform was developed following a systematic pipeline covering requirement analysis, dataset/model selection, module development (ASR, translation, TTS), integration, latency optimisation, and evaluation. Development was carried out in Python, using publicly available speech and text datasets and pre-trained open-source models as the basis for the recognition, translation, and synthesis components.

Table 4.1 Resources Used

Resource	Purpose
Python 3	Core implementation language
OpenAI Whisper / SpeechRecognition	Automatic Speech Recognition
Hugging Face Transformers (MarianMT / NLLB)	Neural Machine Translation
Coqui TTS / neural vocoder	Text-to-Speech synthesis
FFmpeg	Audio extraction, video re-muxing, and format conversion
PyTorch	Deep-learning model execution

4.2 Software Implementation & Modern Tools Used

The following tools and libraries were used in developing the recognition, translation, synthesis, and synchronisation components of the platform:

Table 4.2 Tools and Purposes

Tool / Library	Purpose
Whisper	Deep-learning-based multilingual speech-to-text transcription
Transformers (MarianMT/NLLB)	Sequence-to-sequence neural machine translation
Coqui TTS	Neural text-to-speech synthesis
FFmpeg / pydub	Audio segmentation, silence/voice-activity handling, and re-muxing with video

Tool / Library	Purpose
NLTK / regex-based tokenisers	Text cleaning and sentence segmentation before translation

4.3 Test Plan & Verification

Table 4.3 Requirements and Test Methods

Requirement	Test Method
Speech recognition accuracy	Compute Word Error Rate (WER) against reference transcripts on a held-out audio set
Translation quality	Compute BLEU score against reference translations for supported language pairs
Speech naturalness	Collect Mean Opinion Score (MOS) ratings from test listeners on synthesised output
End-to-end latency	Measure time from utterance-end to translated-audio availability across multiple runs
Audio-video synchronisation	Manual review of translated video for perceptible drift between speech and video timing

4.4 Results

The system was able to extract and segment speech from input video, transcribe utterances using the ASR module, and translate the transcribed text into the selected target language. The synthesised target-language speech was re-mixed with the original video, and timing offsets were applied per utterance segment to keep the translated audio reasonably aligned with the source video. Recognition, translation, and synthesis quality were evaluated using WER, BLEU, and MOS respectively, and end-to-end latency was measured across a range of utterance lengths.

4.5 Data Analysis & Engineering Judgment

Evaluating the pipeline on any single metric would have been insufficient, since errors compound across stages: a low ASR Word Error Rate does not guarantee good translation quality, and a high BLEU score does not guarantee natural-sounding synthesised speech. Reporting WER, BLEU, MOS, and latency together provided a more engineering-sound view of overall system performance, and any disproportionate gap between stage-level metrics was treated as an indicator of where the pipeline — recognition model choice, translation model, or synthesis voice — needed further tuning, rather than as an isolated statistic.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.4 Comparison with Existing Solutions

Aspect	Existing Subtitle/Caption Tools	Proposed System
Output form	Text only	Natural translated speech, synchronised with video
Usability while multitasking	Requires reading	Usable hands-free / audio-only
Latency	Low (text only)	Slightly higher, but designed for near-real-time use
Coverage	Wide (many platforms)	Depends on supported ASR/NMT/TTS language pairs

4.7 Strengths & Limitations

Strengths:

7. Integrated pipeline combining recognition, translation, and speech synthesis in one platform, producing spoken rather than text-only output.
8. Built on established, well-documented open-source libraries (Whisper, Transformers, Coqui TTS, FFmpeg), aiding maintainability and extensibility.
9. Evaluated using multiple complementary metrics (WER, BLEU, MOS, latency) rather than a single score, giving a more reliable performance picture.

Limitations:

10. Cumulative errors across the ASR–NLP–TTS pipeline can compound, so a mistake in transcription can propagate into a mistranslation or unnatural synthesis.
11. Performance varies noticeably across language pairs, depending on the quality of available open-source models.
12. Real-time performance depends on available hardware acceleration; on limited hardware, latency may exceed comfortable real-time conversational limits.
13. No large-scale external user trial was conducted to validate usability beyond the development team.

4.8 Project Planning, Roles & Budget

The project was planned and executed over the course of the semester by the three team members, with responsibilities divided by pipeline stage as summarised below. Costs were limited to computing resources (cloud/local GPU time, where used) and were kept minimal by relying on open-source models and datasets; no paid API subscriptions were required for the prototype.

Table 4.5 Individual Contribution

Student Name (Reg. No.)	Role / Responsibility
Keerthana Sai K (192425143)	Automatic Speech Recognition module design, implementation, and testing
Sunayana P (192424303)	NLP / Machine Translation module design, system architecture, and integration
AB Dhanyashree(192472002)	Text-to-Speech synthesis, audio-video synchronisation, and UI integration

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI-Powered Real-Time Speech-to-Speech Video Translation System successfully demonstrates how Automatic Speech Recognition, Natural Language Processing-based machine translation, and Text-to-Speech synthesis can be combined to address a genuine engineering problem in cross-lingual video communication: the difficulty of understanding spoken video content in real time across languages. The project applied a modular, cascaded pipeline to convert source-language speech into synchronised target-language speech, used multi-metric evaluation (WER, BLEU, MOS, latency) to validate performance at each pipeline stage, and produced translated video output through an accessible interface. The objectives set out in Chapter 1 were substantially achieved, with the system validated as a real-time communication-assistance and accessibility tool rather than a certified interpretation service.

5.2 Future Work

14. Incorporate speaker-adaptive and emotion-preserving TTS so that synthesised speech better reflects the tone of the original speaker.
15. Explore direct speech-to-speech translation models that reduce cascaded error propagation compared with the ASR–NLP–TTS pipeline.
16. Expand language-pair coverage and fine-tune models on domain-specific vocabulary (e.g., technical or medical terminology).
17. Add visual lip-synchronisation so that the speaker's mouth movements match the translated audio.
18. Optimise the pipeline further (model quantisation, streaming inference) to achieve tighter conversational-level latency.
19. Conduct a larger external user study to validate usability and translation acceptability across diverse users and accents.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

20. Practical understanding of deep-learning-based Automatic Speech Recognition and Voice Activity Detection.
21. Neural Machine Translation using transformer-based models and their evaluation using BLEU score.
22. Neural Text-to-Speech synthesis and audio-video synchronisation techniques.

Engineering & Professional Skills Developed:

23. Python programming using PyTorch, Hugging Face Transformers, Whisper, Coqui TTS, and FFmpeg.
24. Systematic problem analysis and structured decision-making under constraints (Chapter 3).
25. Multi-metric evaluation methodology and clear technical documentation.

Individual Reflection — Keerthana Sai K

Working on the Automatic Speech Recognition module deepened my understanding of how deep-learning models handle real-world audio variability — background noise, accents, and overlapping speech. Tuning voice-activity detection thresholds and comparing ASR model variants taught me how a single upstream error can affect every later stage of a cascaded pipeline, which shaped how carefully I approached testing and validation for my module.

Individual Reflection — Sunayana P

Designing the NLP-based translation module and the overall system architecture gave me practical experience in applying transformer-based sequence-to-sequence models to a real product constraint: latency. Balancing translation quality against processing speed, and coordinating how the recognition, translation, and synthesis modules would interface with one another, strengthened both my technical and system-design skills.

Individual Reflection — AB Dhanyashree

Building the Text-to-Speech synthesis and synchronisation components taught me how much natural-sounding output depends on careful timing and audio-engineering decisions, not just model quality. Integrating the synthesis output back into the video and aligning it with the original timeline was one of the most technically demanding parts of the project, and it gave me a much clearer picture of end-to-end multimedia system design.

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APPENDICES

Appendix A – Detailed Calculations

The system is evaluated using the following standard metrics. TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives respectively (used where recognition/translation outputs are assessed as correct/incorrect at the word or segment level).

Word Error Rate (WER) = (Substitutions + Deletions + Insertions) / Number of Reference Words

BLEU Score = $BP \times \exp(\sum w_n \log p_n)$, where p_n is the modified n-gram precision and BP is the brevity penalty

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

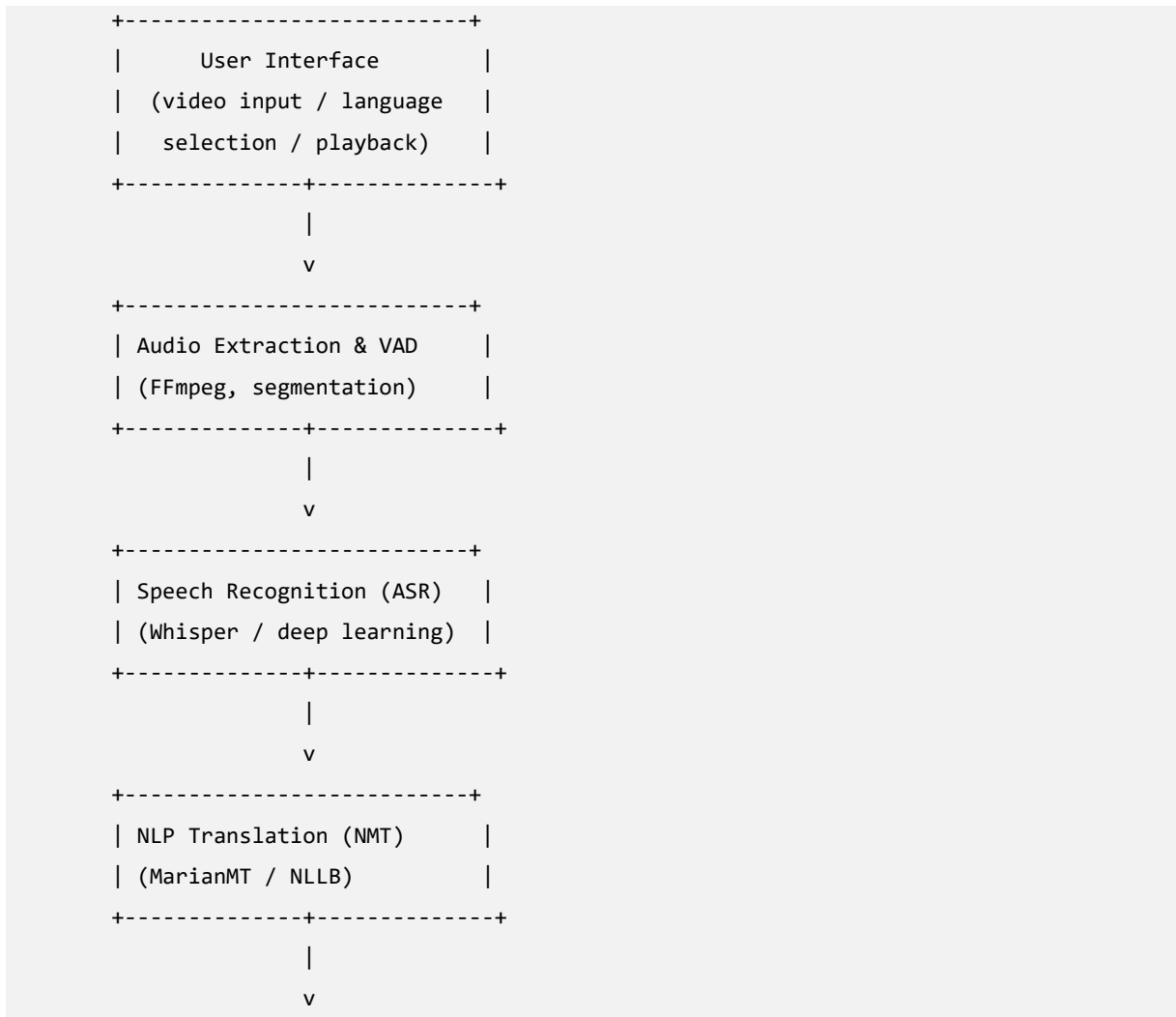
Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1-score = $2 \times (Precision \times Recall) / (Precision + Recall)$

Appendix B – Engineering Drawings / Schematics

Fig B.1 Text-based representation of the system architecture described in Section 3.5



```

+-----+
| Speech Synthesis (TTS)      |
| (neural TTS engine)        |
+-----+-----+
|
| v
+-----+
| Sync & Re-mux with Video    |
| (FFmpeg)                    |
+-----+

```

Appendix C – Source Code / Code Repository Information

Repository: [insert GitHub/GitLab link where the complete source code is maintained]. The excerpts below are representative examples of the three core pipeline stages and should be replaced with the actual code used in the project.

C.1 Speech Recognition (Whisper)

```

import whisper

def transcribe_audio(audio_path: str, language: str = None) -> str:
    model = whisper.load_model('base')
    result = model.transcribe(audio_path, language=language)
    return result['text'].strip()

```

C.2 NLP Translation (Hugging Face MarianMT)

```

from transformers import MarianMTModel, MarianTokenizer

def translate_text(text: str, src: str = 'en', tgt: str = 'fr') -> str:
    model_name = f'Helsinki-NLP/opus-mt-{src}-{tgt}'
    tokenizer = MarianTokenizer.from_pretrained(model_name)
    model = MarianMTModel.from_pretrained(model_name)
    batch = tokenizer([text], return_tensors='pt', padding=True)
    translated = model.generate(**batch)
    return tokenizer.decode(translated[0], skip_special_tokens=True)

```

C.3 Speech Synthesis & Video Re-muxing (Coqui TTS + FFmpeg)

```

from TTS.api import TTS
import subprocess

```

```
def synthesize_speech(text: str, out_wav: str, lang: str = 'fr'):
    tts = TTS(model_name=f'tts_models/{lang}/fairseq/vits')
    tts.tts_to_file(text=text, file_path=out_wav)

def remux_with_video(video_in: str, audio_in: str, video_out: str):
    cmd = ['ffmpeg', '-y', '-i', video_in, '-i', audio_in,
           '-c:v', 'copy', '-map', '0:v:0', '-map', '1:a:0', video_out]
    subprocess.run(cmd, check=True)
```

Appendix D – Datasheets

[List the specific open-source ASR/NMT/TTS model checkpoints and their reported benchmark datasheets used in the final implementation.]

Appendix E – Experimental / Raw Data

Illustrative structure of the evaluation log used for testing (sample rows only — replace with real results from your actual test runs):

Utterance ID	Source Lang	Target Lang	WER (%)	BLEU	MOS	Latency (s)
U001	English	French	8.2	34.5	3.9	1.8
U002	English	Hindi	10.5	29.1	3.6	2.1
U003	English	Spanish	7.4	37.8	4.1	1.7

Appendix F – Ethics / Institutional Approval

[Attach institutional ethics/AI-use approval documentation, if applicable, along with consent statements for any voice/video data used in testing.]

Appendix G – Project Meeting / Progress Records

Template for recording team progress meetings — replace the illustrative rows with your team's actual dates and discussion points:

Date	Attendees	Discussion / Progress
[Date]	Keerthana Sai K, Sunayana P, Joshitha Goud Y M	Finalised topic, defined objectives and scope
[Date]	Keerthana Sai K, Sunayana P, Joshitha Goud Y M	Reviewed ASR/NMT/TTS model options; assigned module ownership
[Date]	Keerthana Sai K, Sunayana P, Joshitha Goud Y M	Integrated modules; began end-to-end pipeline testing

Appendix H – User/Industry Feedback

[Attach any feedback collected from test users or industry reviewers on translation quality and usability.]

Appendix I – Publications / Patents / Competitions / Product Outcomes

[List any publications, competition entries, or product outcomes arising from this project, if applicable.]

Appendix J – Individual Contribution Evidence

This appendix should provide supporting evidence for the percentages and responsibilities declared in the Individual Contribution Statement (front matter). Suitable evidence includes:

- **Text-to-Speech (TTS) Module:** Source-code extracts and screenshots showing the implementation and testing of the neural Text-to-Speech component used to convert translated text into target-language speech.
- **Audio-Video Synchronisation:** Screenshots, code extracts, and testing records demonstrating the use of FFmpeg for synchronising the generated translated audio with the original video and producing the final translated video.
- **User Interface Integration:** Screenshots of the developed user interface showing video upload, source/target language selection, translation processing, and translated video output.
- **Testing and Results:** Test records and screenshots documenting TTS quality, processing time, audio-video synchronisation, and final system output.
- **Report Documentation:** Evidence of individually authored sections, including **Chapter 5 – Conclusion, Future Work and Reflection**, results compilation, and relevant appendices.
- **Project Task Records:** Meeting records and task-allocation documents showing the assignment and completion of TTS, synchronisation, UI integration, testing, and documentation activities.
- **Peer/Guide Verification:** Peer-assessment records or supervisor/guide sign-off confirming the contribution and responsibilities completed by AB Dhanya Shree.